



PRESSURIZED ROVER PROCEDURES

Relevant to: PR Teams

NASA SUITS
2025-26 Challenge

NASA SUITS Pressurized Rover Procedures

Last Updated: 3/2/2026

Updated By: Tech Team

This document **may** be updated ahead of test week in May 2026.

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Notes:

This document will likely undergo several more revisions as we do dry runs in preparation for test week. NASA SUITS aims to have each test session run for 45-minutes. This allows time on each end of the window to setup and to transition from group to group. Therefore, the PR and Spacesuit portions will each take about 25-minutes with a few cutoffs. The reasoning for the cutoffs is to prevent teams from getting stuck and allowing other functionalities to be showcased, i.e. “Something isn’t working, and we are behind, so in order to see your other functionalities, let’s move on.”

Change Log:

1. Modify PR Team timeline to be more real-world accurate. Add verbal aspects to checklists for design evaluators to confirm procedures, add cover page. -RE 2-17-26
2. Separate PR and EV team procedures into separate documents. -RE 2-18-26
3. Fix discrepancy in timeline table vs. Procedures list. -RE 2-19-26
4. Alter procedures to reflect Test Week more accurately, change wording to be uniform throughout -GC, LC, RE 2-27-26
5. Add procedures for handling errors that may occur -LC, RE 3-2-26

General Test Timeline

Both the rover and EV teams will begin their procedures at the same time. The following procedures are done concurrently with the EV team and their procedures but are **NOT** dependent on one another. Cutoff times are fluid for each station, but 45 min is hard stop.

PR	Ideal start time	Time to complete
Pre-Nav Checklist	0	3 min
Navigate to LTV last known location	3 min	3 min
Initial ping and calculate search area	6 min	2 min
Execute search procedure, find LTV	8 min	10 min
Navigate back to HAB (can also showcase EVA monitoring functionality during this time)	18 min	4 min
Showcase any remaining EVA monitoring functionality	22 min	3 min
Conclusion of the test	25 min	NA

Procedures

CAPCOM:

- *START Rover Telemetry*
- *Announce scenario start, give go to begin*

Pressurized Rover

Complete Pre-Navigation Checklist (3 min)

1. Pilot verbally confirm battery level is > 95%
2. Pilot verbally confirm O2 levels are > 95%
3. Pilot verbally confirm O2 pressure is > 2900 psi
4. Pilot verbally confirm PR Cabin Pressure is > 3.95 psi

5. Pilot verify PR headlights are operational by manually cycling lights ON, OFF, ON, OFF, ON, verbally confirm success
6. Pilot drop pin at current location, verbally confirm success
7. Drop pin at LTV last known position, verbally confirm success
8. Calculate optimal path to the LTV last nominal position
9. Verbally announce completion of checklist

Navigate to LTV last known location (3 min)

1. Begin navigation to LTV last nominal position
 - a. *Demonstrate pathfinding, autonomous navigation, and hazard avoidance capability*
 - b. *It is highly recommended that teams include the ability to manually pilot the PR*
2. Upon arrival, fully stop the PR, verbally confirm success
3. Verbally announce arrival at LTV last nominal position
4. Check telemetry data and look for any off-nominal values before proceeding
5. Verbally announce beginning LTV search

Initial ping and calculate search area (2 min)

1. Send first ping, analyze the incoming RSSI (signal strength)
2. Calculate a search area based on the LTV max speed and time away from last nominal position
3. Set pin for next ping location and calculate optimal path of navigation
4. Review consumables, verify and verbally confirm point-of-no-return state is nominal
 - a. If PNR is off nominal, begin return to HAB and disregard further procedures

Execute search procedure, find LTV (10 mins)

1. Begin navigation to next ping location
2. Upon arrival, fully stop the PR, verbally confirm success
3. Send ping and wait for new RSSI value, verbally confirm new RSSI update
4. Determine next ping location based on RSSI
5. Repeat steps 1-4 until LTV is located
6. Once LTV is located, verbally confirm visual of LTV
7. Drop pin at exact location of LTV, verbally confirm success

Navigate back to HAB (4 min)

1. Calculate the optimal path back to HAB using the provided map of the DUST lunar environment
2. Navigate back to HAB
 - a. *Demonstrate pathfinding, autonomous navigation, and hazard avoidance capability*
 - b. *It is highly recommended that teams include the ability to manually pilot the PR*
8. Upon arrival at the HAB, fully stop the PR, verbally confirm success
3. Announce arrival at HAB and conclusion of EVA

Showcase Design Features (3 min)

1. Teams may use remaining time to showcase EVA monitoring functionality
2. Teams may produce a custom set of procedures to showcase various elements of design such as:
 - a. Further examples of resource utilization predictive modeling
 - b. Monitor EVA video feed (if applicable)
 - c. Caution and warning features

CAPCOM

- *Stop Rover Telemetry*

Handling Error Scenarios

If an error occurs, the relevant procedures may be used to mitigate risk or abort the EVA. Please note that the occurrence of one or more of the following errors should only be used to test your design's response and will NOT prematurely end your test if mishandled.

Off Nominal Cabin Temperature: Below Nominal Range

1. Cabin Heating – ON
2. Cabin Cooling – OFF
 - a. *These will be sent via UDP command to the TSS*

Off Nominal Cabin Temperature: Exceeding Nominal Range

1. Cabin Heating – OFF

2. Cabin Cooling – ON

a. These will be sent via UDP command to the TSS

Too dark to see

1. Headlights – ON

a. This will be sent via UDP command to the TSS